

OFDMA Resource Block Allocation

1. Aim

To simulate and analyze OFDMA (Orthogonal Frequency Division Multiple Access) resource block allocation and evaluate system performance in terms of resource utilization and user throughput using MATLAB.

2. Software Used

MATLAB

3. Code

```
clc;
clear;
close all;

% Number of Users
numUsers = 4;

% Number of Resource Blocks
numRB = 12;

% Resource Block Allocation Matrix
allocation = zeros(numUsers, numRB);

% Allocate RBs in Round Robin
for rb = 1:numRB
    user = mod(rb-1, numUsers) + 1;
    allocation(user, rb) = 1;
end

% Display Allocation Matrix
disp('OFDMA Resource Block Allocation Matrix');
disp(allocation);

% Plot Allocation
figure;
imagesc(allocation);
```

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end

% Display Allocation Matrix

disp('OFDMA Resource Block Allocation Matrix');

disp(allocation);

% Plot Allocation

figure;

imagesc(allocation);

colormap(jet);

colorbar;

xlabel('Resource Blocks');

ylabel('Users');

title('OFDMA Resource Block Allocation');

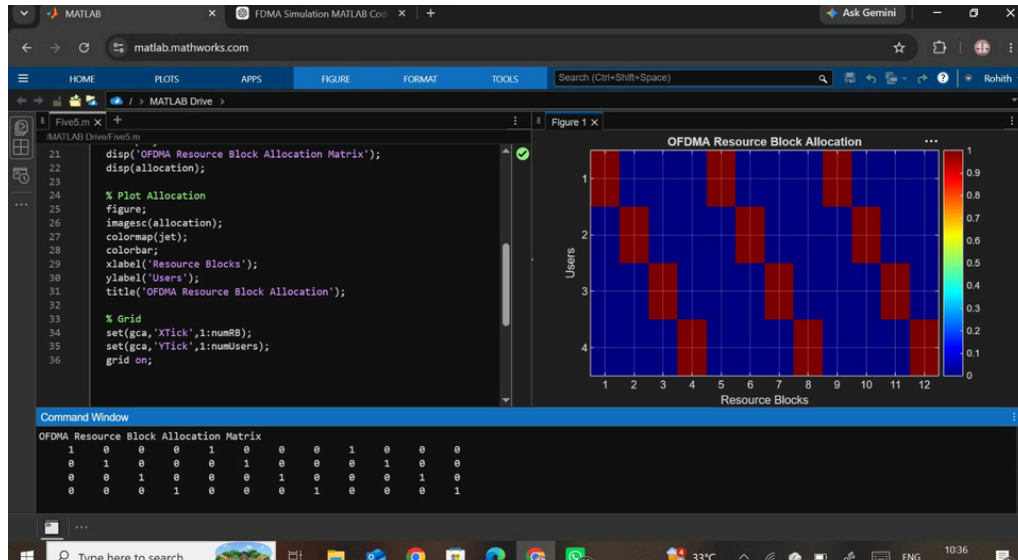
% Grid

set(gca,'XTick',1:numRB);

set(gca,'YTick',1:numUsers);

grid on;

6. Output Graphs



7. Result

The OFDMA Resource Block Allocation was successfully implemented using MATLAB. Resource Blocks (RBs) were allocated to multiple users using a round-robin scheduling technique. The allocation matrix and graphical output clearly show the distribution of resource blocks among users, demonstrating efficient spectrum utilization and fair allocation in an OFDMA-based communication system.
